

Notes: Bollen & Pool (JF 2009)

Do Hedge Fund Managers Misreport Returns? Evidence from the Pooled Distribution

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1 What the paper does and why it matters

1.1 Research question

The paper asks whether some hedge fund managers *misreport* returns, in a systematic way, by *avoiding the appearance of small losses*. The empirical signature is a **discontinuity at zero** in the pooled distribution of reported monthly hedge-fund returns: too few small negative returns and too many small positive returns.

1.2 Core idea

If reported returns are occasionally distorted (e.g., by discretionary valuation of illiquid positions), then:

1. The **pooled histogram** of reported returns should show “missing mass” just below 0 and “extra mass” just above 0.
2. If distortions are temporary, they must be **reversed**, implying weaker subsequent performance for funds that exhibit strong discontinuities.
3. Discontinuities should be **less pronounced when monitoring is tighter** (e.g., around audits).
4. Discontinuities should be **more pronounced when discretion is feasible**, e.g., for illiquid strategies / more return smoothing.

1.3 Main findings

- A robust discontinuity exists at 0 for hedge-fund monthly returns (Figures 1–3; Table I).
- Investors’ flows respond strongly to the *frequency of positive months*, creating incentives to avoid reporting losses (Table II).
- Evidence consistent with reversals: the discontinuity disappears in bimonthly returns; subsequent returns behave as if small positives contain inflated observations; funds with “kinked” distributions suffer out-of-sample deterioration (Figure 4; Tables III–IV).
- Alternative explanations are weakened by a battery of tests: the discontinuity is absent near audits, absent in fitted returns from market-based factor payoffs, absent for CTAs, and strongest for illiquid/smoothed funds (Tables V–VI; Figure 5).
- Economic magnitude: the paper estimates large wealth transfers from incoming to existing investors when inflows occur at inflated NAVs (Table VII).

2 Data and objects of interest

2.1 Return data

- Hedge fund monthly returns (CISDM database¹), sample period January 1994–December 2005.
- For comparison: Commodity Trading Advisors (CTAs) and, as robustness, CRSP equity mutual funds and CRSP stock returns.
- The paper removes returns reported as exactly 0.0000 and consecutive 0.0001 observations (these could reflect missing data or manipulation).

2.2 Why pooled distributions?

The key statistical object is the **pooled cross-sectional time-series distribution** of monthly returns:

$$\{R_{i,t}\}_{i=1,\dots,N_t; t=1,\dots,T}.$$

Pooling creates enormous sample size, enabling sharp inference about the density near economically meaningful thresholds (here, 0).

3 Empirical methodology: testing for discontinuities in the pooled return distribution

3.1 Step 1: choose a histogram bin width

A histogram is a nonparametric density estimator whose statistical properties depend heavily on the **bin width**. The paper follows Silverman (1986) and sets the bin width to

$$\text{BinWidth} = \alpha \times 1.364 \min\left(\sigma, \frac{Q}{1.340}\right) N^{-1/5}, \quad (1)$$

where σ is the sample standard deviation, Q is the interquartile range, N is the number of observations, and α is a scale factor depending on distributional assumptions. In implementation, the paper sets $\alpha = 0.776$ (corresponding to normality).

Interpretation. Equation (1) balances variance and bias of the histogram density estimator. Figure 2 shows why: too-small bins create “noisy” local irregularities; too-large bins hide local structure.

¹For an overview of the CISDM databases, see <https://www.isenberg.umass.edu/centers/center-for-international-securities-and-derivatives-markets/cisdms-databases>

3.2 Step 2: estimate a smooth reference density under the null (kernel density)

The null hypothesis is *no discontinuities* (smooth density). The paper uses a Gaussian kernel density estimate:

$$\hat{f}(t; h) = \frac{1}{Nh} \sum_{i=1}^N \varphi\left(\frac{x_i - t}{h}\right), \quad (2)$$

where h is the bandwidth, $\{x_i\}$ are the observed returns, and $\varphi(\cdot)$ is the standard normal density.

Why nonparametric? A parametric density may fit poorly (fat tails, skewness, multi-modality). A kernel density can match “global” shape while still treating any local discontinuity as a deviation from the smooth null.

3.3 Step 3: compute bin-by-bin test statistics comparing observed vs expected counts

Let a histogram bin have observed count x out of total n . Let p be the probability that an observation falls in that bin under the smooth reference density (from the kernel estimate). Under a binomial approximation, the paper uses the standard normal statistic

$$Z = \frac{x - np}{\sqrt{np(1 - p)}}. \quad (3)$$

Under the null of a smooth distribution, Z is approximately standard normal.

Reading the test. Large negative Z means “too few” observations in that bin relative to the smooth null; large positive Z means “too many.” The signature of loss-avoidance manipulation is a **negative** Z in the bin just below zero and a **positive** Z in the bin just above zero.

3.4 Step 4: simulation-based confidence bands (smoothed bootstrap)

As an additional robustness check, the paper uses a smoothed bootstrap (following Hörmann and Leydold (2000)):

1. Draw observations from the original data with replacement.
2. Add kernel noise to create a synthetic sample consistent with the fitted smooth density.
3. For each synthetic sample, compute the histogram bin counts.
4. Use the empirical distribution across simulations to form confidence bands for bin heights.

Figure 3 visualizes the discontinuity by overlaying the KDE-implied bin heights and 99% simulated confidence bands.

4 Why would managers avoid small losses?

4.1 Investor flow-performance relation

The paper argues that if investors react nonlinearly to losses (or care about the frequency of positive months), managers have incentives to avoid reporting negative returns.

4.1.1 Model (annual flows)

The empirical specification is:

$$F_{i,t} = \alpha_1 + \alpha_2 Y_{i,t-1} + (\beta_1 + \beta_2 Y_{i,t-1}) R_{i,t-1} + (\beta_3 + \beta_4 Y_{i,t-1}) \text{NPOS}_{i,t-1} + \varepsilon_{i,t}, \quad (4)$$

where:

- $F_{i,t}$ is percentage fund flow in year t .
- $Y_{i,t-1}$ is an indicator for “young” funds (age ≤ 3 years) in year $t - 1$.
- $R_{i,t-1}$ is cumulative annual return in year $t - 1$.
- $\text{NPOS}_{i,t-1}$ is the number of months with nonnegative returns in year $t - 1$.

Fund flow is computed from total net assets (TNA):

$$F_{i,t} = \frac{\text{TNA}_{i,t} - \text{TNA}_{i,t-1}(1 + R_{i,t})}{\text{TNA}_{i,t-1}}. \quad (5)$$

The paper also considers variants replacing NPOS with the number of months beating the S&P 500 (NSP) and including both (Table II).

4.1.2 Table II interpretation

- **Panel A (NPOS):** flows increase with lagged annual return ($\beta_1 > 0$) *and* increase strongly with the number of nonnegative months ($\beta_3, \beta_4 > 0$).
- The magnitudes imply that investors place heavy weight on “not losing money often,” consistent with a behavioral or institutional preference for steady positive returns.
- **Panel B (NSP):** flows also respond to the number of months beating the market.
- **Panel C (both):** NPOS remains strongly significant even controlling for NSP, suggesting NPOS contains information investors value beyond relative performance.

Bottom line: because flows react to the incidence of nonnegative months, managers have incentives to avoid reporting small monthly losses.

Parameter	Estimate	<i>t</i> -Statistic
Panel A: Number of Returns ≥ 0		
β_1	0.1506	2.0201*
β_2	-0.0306	-0.2921
β_3	0.0665	10.1794**
β_4	0.0540	4.9511**
Adjusted R^2	0.0857	
Panel B: Number of Returns $\geq$ S&P 500		
β_1	0.0722	1.3087
β_2	0.0120	0.1430
β_3	0.0805	9.7051**
β_4	0.0311	2.7324**
Adjusted R^2	0.0761	
Panel C: Both		
β_1	0.0241	0.4463
β_2	0.0105	0.1470
β_3	0.0524	8.3311**
β_4	0.0487	4.7794**
β_5	0.0525	6.2865**
β_6	0.0213	1.9541
Adjusted R^2	0.0918	

Table II Flow-Performance Relation

5 Main results: evidence consistent with misreporting

5.1 Pervasiveness over time (Table I)

Table I reports the discontinuity test statistics by year for (i) equity mutual funds and (ii) hedge funds.

- For hedge funds, the bin *below* zero has strongly negative Z statistics each year (missing small losses), while the bin *above* zero has strongly positive Z (excess small gains).
- For mutual funds, the bins around zero are mostly not abnormal, and crucially do not show a

Year	Mutual Funds		Hedge Funds	
	Negative	Positive	Negative	Positive
1994	-2.16	0.58	-2.92	5.64
1995	-0.81	-0.17	-4.53	2.98
1996	-1.57	-0.28	-5.19	0.84
1997	-1.64	0.12	-6.90	4.87
1998	-0.98	2.14	-6.59	9.24
1999	-1.66	-0.50	-6.41	7.21
2000	1.03	0.11	-6.67	4.15
2001	-1.61	1.33	-6.11	10.29
2002	2.34	1.56	-4.94	8.11
2003	-4.25	-3.87	-5.49	8.07
2004	-4.40	-1.10	-5.48	4.30
2005	-0.94	-0.24	-4.56	3.57

Table I Discontinuity by Year

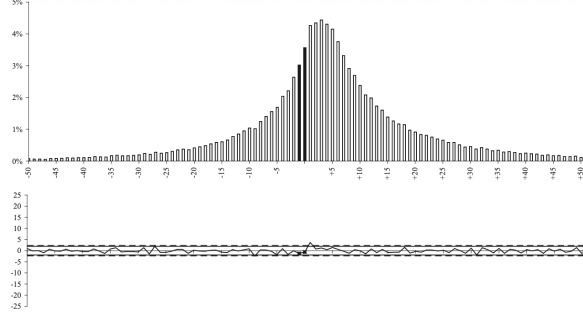


Figure 4. Bimonthly returns. The figure includes bimonthly returns of all hedge funds in the 2005 CISDM database, $n = 211,908$. The top graph is a histogram of returns. Bold vertical bars indicate returns bracketing zero. Bins are 36 basis points wide with 101 bins displayed, centered

discrete jump at zero.

Interpretation: the discontinuity is not a generic feature of professionally managed pooled returns; it appears specific to settings where managers have more discretion in valuation and reporting.

5.2 Reversals test 1: bimonthly returns (Figure 4)

If small monthly gains are inflated and reversed quickly, then aggregation to bimonthly returns can eliminate the discontinuity. The paper forms nonoverlapping bimonthly returns and finds that the distribution is smooth (Figure 4): no bins reject the smooth null.

Interpretation: this is consistent with short-horizon reversals of overstatements.

5.3 Reversals test 2: conditional subsequent performance near zero

The paper compares cumulative subsequent returns following observations in the bin just below zero (“Negative”) versus just above zero (“Positive”). Under positive serial correlation, Positive should predict higher subsequent returns. Instead, subsequent performance following the Positive bin is lower than that following the Negative bin over horizons of 1–6 months (reported as statistically significant for several specifications).

Interpretation: small positives behave as if they contain reversable inflation.

5.4 Reversals test 3: out-of-sample deterioration for “Offender” funds (Table III)

5.4.1 Defining the kink measure

For each fund, define

$$\text{Kink}_i \equiv \Pr_i(0 \leq R < 50 \text{ bp}) - \Pr_i(-50 \text{ bp} \leq R < 0), \quad (6)$$

measured using the fund’s empirical return distribution. Funds are classified as **Offenders** if the mass in $[0, 50\text{bp}]$ is significantly greater (at 5%) than the mass in $[-50\text{bp}, 0]$; otherwise they are **Non-offenders**.

No. of Funds		μ	σ	<i>Sharpe</i>	<i>Skew</i>	<i>Kurt</i>	<i>Kink</i>
A. Live Funds							
A1. Non-offenders							
1,258	<i>Pre</i>	0.0136	0.0426	0.3857	0.1147	1.9715	0.0234
	<i>Post</i>	0.0103	0.0311	0.3227	0.0171	1.2621	0.0377
	<i>p-value</i>	0.0000	0.0000	0.0005	0.0110	0.0000	0.0001
A2. Offenders							
185	<i>Pre</i>	0.0101	0.0188	0.6120	−0.0798	3.6191	0.2356
	<i>Post</i>	0.0067	0.0142	0.3447	−0.1029	1.8805	0.1701
	<i>p-value</i>	0.0000	0.0200	0.0000	0.8633	0.0024	0.0016
Difference		0.9260	0.0000	0.0000	0.5487	0.0586	0.0000
B. Dead Funds							
B1. Non-offenders							
1,478	<i>Pre</i>	0.0148	0.0508	0.2998	0.0576	1.7801	0.0223
	<i>Post</i>	0.0038	0.0546	0.0416	−0.1083	1.8605	0.0426
	<i>p-value</i>	0.0000	0.0313	0.0000	0.0000	0.5305	0.0000
B2. Offenders							
144	<i>Pre</i>	0.0133	0.0386	0.3322	0.2287	3.6085	0.2441
	<i>Post</i>	0.0036	0.0237	−0.1206	−0.1657	3.0947	0.1285
	<i>p-value</i>	0.0081	0.3177	0.0001	0.0183	0.3531	0.0000
Difference		0.7006	0.2138	0.0573	0.1230	0.2771	0.0000

Table III Reversals

5.4.2 Table III interpretation

Table III splits each fund’s history into halves (Pre, Post) and reports cross-sectional means of summary statistics. Key patterns:

- **Live funds:** Offenders exhibit a large drop in Sharpe ratio from Pre to Post, substantially larger than Non-offenders. This is consistent with an initial “inflation” that later must unwind.
- Offenders exhibit more negative skewness and higher excess kurtosis than Non-offenders, consistent with replacing small negatives by small positives and occasionally “catching up” with larger negative months.
- **Dead funds:** both groups deteriorate, but Offenders still show sharp drops in Sharpe and large changes in Kink.

5.5 Persistence and smoothing: does Kink add information beyond AR(1) smoothing? (Table IV)

To separate “kink” from generic smoothing/serial correlation, the paper estimates the cross-sectional regression:

$$\text{Sharpe}_{i,\text{Post}} = \alpha + \beta_0 \text{Sharpe}_{i,\text{Pre}} + \beta_1 I_{i,\text{Pre}} \text{Sharpe}_{i,\text{Pre}} + \varepsilon_i, \quad (7)$$

where $I_{i,\text{Pre}}$ is an indicator either for (i) Offender status as defined by Kink, or (ii) a significantly positive AR(1) coefficient in Pre.

		α		β_0	β_1
A. Sorted by Kink					
		A1. Live Funds			
Adjusted R^2	29.0%	Estimate	0.1223	0.5147	-0.1306
No. of observations	1443	p-value	0.0000	0.0000	0.0008
		A2. Dead Funds			
Adjusted R^2	6.0%	Estimate	-0.0601	0.2599	0.2931
No. of observation	1622	p-value	0.0001	0.0000	0.0011
B. Sorted by AR(1)					
		B1. Live Funds			
Adjusted R^2	28.5%	Estimate	0.1248	0.4725	0.0335
No. of observation	1443	p-value	0.0000	0.0000	0.2846
		B2. Dead Funds			
Adjusted R^2	5.5%	Estimate	-0.0570	0.2972	-0.0712
No. of observation	1622	p-value	0.0002	0.0000	0.1782

Table IV Persistence

Table IV interpretation.

- **Panel A (Kink indicator):** for live funds, Sharpe ratios are persistent ($\beta_0 > 0$), but Offenders have *significantly lower persistence* (negative β_1). This implies that conditional on initial Sharpe, Offenders underperform in the second half.
- **Panel B (AR(1) indicator):** the AR(1)-based indicator does not materially predict Sharpe persistence (insignificant β_1). Thus, Kink is not just capturing unconditional smoothing.

Bottom line: Kink contains incremental information about subsequent performance deterioration consistent with misreporting.

6 Alternative explanations and robustness (Tables V–VI; Figure 5)

6.1 Audit oversight (Table V and Table VI: Audit vs No Audit)

Idea. If the discontinuity is due to managerial skill, it should not disappear around audits. If it is due to discretionary reporting, tighter monitoring around audits should shrink the discontinuity.

Evidence.

- Table VI shows the return distribution **around audits** (audit month and two months prior) has no significant discontinuity at zero.
- A matched sample of “no-audit-information” funds exhibits a strong discontinuity.
- Table V shows audited funds are older and larger on average, and a probit suggests audited status is related to fund attributes (age, size, active status, etc.).

Panel A: Univariate Statistics			
Attribute	Audited	Non-audited	<i>t</i> -Statistic
Monthly return	1.06%	0.74%	5.3643**
Size (in \$000)	192,922	95,516	1.2839
Age	5.04	2.47	26.1894**
Panel B: Probit Regression Coefficients			
Parameter	Estimate	<i>z</i> -Statistic	
Constant	-1.1419	-4.4752**	
Active	0.3557	7.2033**	
Age	0.1636	14.8648**	
Ln(size)	0.0485	3.1606**	
Onshore	0.1077	2.3485*	
Monthly return	4.9241	3.3834**	
McFadden R^2	0.1353		

Table V Audits and Fund Attributes

Bin	Audit	No Audit	Fitted	Raw	Non-put Sellers	Put Sellers	Equity Neutral	Distressed Securities	Low Smooth	High Smooth	Live	Dead
-5	-1.05	0.18	0.38	-1.74	0.75	-0.10	0.19	0.66	-1.22	-0.98	-0.26	-1.25
-4	-0.20	-2.19	-0.89	-2.51	-2.99	0.46	-0.74	1.62	-1.41	-1.36	-0.57	-0.11
-3	-0.73	-1.14	0.01	1.48	1.60	-1.46	-1.22	-1.18	-0.45	0.13	-2.97	-2.55
-2	-1.09	0.02	-0.07	-3.81	-3.87	-1.92	0.22	-1.67	-0.73	-1.64	-0.12	-2.85
-1	-1.32	-4.07	-0.59	-15.27	-12.24	-10.58	-0.09	-5.00	-2.71	-8.33	-8.61	-10.35
Zero	0.33	3.86	0.12	15.02	13.84	9.13	3.62	3.78	6.73	7.11	8.01	15.03
+1	2.26	1.82	0.73	2.09	-0.30	2.24	-1.24	0.07	0.52	1.74	5.36	2.11
+2	1.51	1.42	0.58	3.05	1.33	0.54	-0.50	1.38	2.04	-0.13	-1.68	-0.57
+3	1.60	0.87	1.85	-0.12	0.69	0.34	2.07	1.18	1.17	0.64	3.77	2.11
+4	-1.22	1.21	0.35	-0.70	2.07	1.02	0.14	2.77	-0.24	3.17	0.31	1.54
+5	-0.67	-0.62	1.01	5.19	1.68	0.47	1.48	-0.78	0.02	0.59	3.77	0.13

Table VI Alternative Explanations

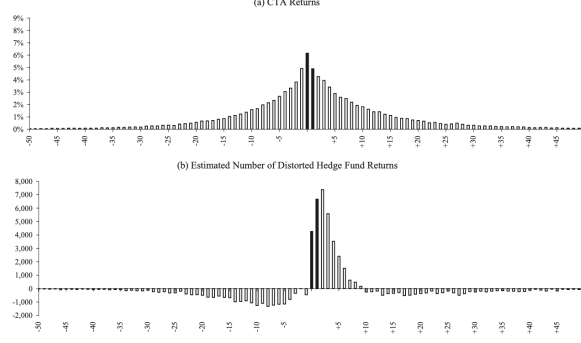


Figure 5. CTA returns. A: A histogram of CTA returns. B: Shows for each bin the estimated number of hedge fund returns that may have been distorted. Estimates are computed by comparing

Interpretation: the discontinuity is sensitive to monitoring intensity, supporting a misreporting mechanism.

6.2 Nonlinear payoffs / dynamic strategies (Table VI: Fitted vs Raw)

Factor model construction. For each fund with at least 24 contiguous observations, estimate excess returns with a linear factor model using up to three factors (selected from a 10-factor set via BIC):

$$R_{i,t} - r_{f,t} = \alpha_i + \beta_i' F_t + u_{i,t}, \quad (8)$$

where F_t includes the three Fama–French factors and additional market-based factors used in the hedge-fund literature. Construct fitted returns by adding back the risk-free rate:

$$\hat{R}_{i,t} = r_{f,t} + \hat{\alpha}_i + \hat{\beta}_i' F_t.$$

Evidence. Table VI shows the pooled distribution of *fitted* returns is smooth (no discontinuity), while the *raw* returns show extreme discontinuities at the bins around zero.

Interpretation: the discontinuity is not an inherent feature of payoffs to commonly-used market-based strategy factors.

6.3 Option-like strategies (Table VI: Put Sellers vs Non-put Sellers)

To test whether discontinuities arise mechanically from selling out-of-the-money puts (many small gains, rare large losses), the paper uses option payoff factors (Agarwal–Naik style) and classifies funds by their exposure to OTM put payoffs.

Evidence: both put sellers and non-put sellers exhibit a discontinuity in Table VI.

Interpretation: put-writing is not the primary driver of the observed kink at zero.

6.4 Liquidity and strategy differences (Table VI; Figure 5)

Two complementary approaches:

1. **Strategy labels:** compare Equity Market Neutral (liquid) vs Distressed Securities (illiquid). Table VI suggests the discontinuity is more consistent with illiquid categories.

2. **Return smoothing:** use the Getmansky–Lo–Makarov (2004) MA(2) framework to estimate a smoothing parameter.

6.4.1 MA(2) smoothing model (Getmansky–Lo–Makarov style)

For each fund with at least 24 monthly returns, demeaned returns satisfy

$$X_{i,t} = \theta_i^0 \eta_{i,t} + \theta_i^1 \eta_{i,t-1} + \theta_i^2 \eta_{i,t-2}, \quad (9)$$

with constraint

$$1 = \theta_i^0 + \theta_i^1 + \theta_i^2. \quad (10)$$

The parameter θ_i^0 is interpreted as the fraction of “true” current returns reported contemporaneously; smaller θ_i^0 corresponds to more smoothing / illiquidity.

Evidence. Table VI shows the discontinuity is stronger among “High Smooth” (more smoothing / more illiquid) funds than among “Low Smooth” funds.

6.5 Database biases (Table VI: Live vs Dead)

Table VI reports that both live and defunct funds exhibit discontinuities at the bins around zero.

Interpretation: survivorship alone cannot explain the kink.

6.6 CTAs as a comparison group (Figure 5)

CTAs report returns based on more liquid instruments and/or with less valuation discretion. Figure 5 shows CTA returns lack the same discontinuity at zero, and the paper uses the CTA distribution to estimate the number of potentially distorted hedge-fund returns in each bin.

7 Economic significance: wealth transfers induced by inflated NAVs (Table VII)

7.1 Flow and wealth transfer accounting

The paper focuses on months when an Offender fund (defined using the full history) has both (i) an inflow and (ii) a small positive reported return (between 0 and 50 bp). Define monthly flows based on reported returns:

$$F_{i,t} = \text{TNA}_{i,t} - \text{TNA}_{i,t-1}(1 + R_{i,t}^R), \quad (11)$$

where $R_{i,t}^R$ is the *reported* return.

Let $\text{NAV}_{i,t}^R$ denote the reported NAV per share and $\text{NAV}_{i,t}^A$ a proxy for the “fair” NAV per share. The wealth transfer from incoming to existing investors is

$$W_{i,t} = \frac{F_{i,t}}{\text{NAV}_{i,t}^R} (\text{NAV}_{i,t}^R - \text{NAV}_{i,t}^A). \quad (12)$$

Table VII
Impact on Fund Flow

Listed are estimates of the wealth transfer between new and existing hedge fund investors when fund inflows occur in funds with discontinuous return distributions. Funds are split into Offenders, for which the percentage of observations between 0 and 50 basis points is, at the 5% level, significantly greater than the percentage of negative observations no less than -50 basis points, and Non-offenders. For each month that an Offender has a fund inflow and a return between 0 and 50 basis points, wealth transfer is estimated as shares purchased times the difference between the reported net asset value and a proxy for the fair net asset value. The fair net asset value equals lagged reported net asset value scaled by a proxy for true return. "Own" is a proxy using a random draw from the fund's negative returns. Other proxies are draws from the negative returns of a matched set of Non-offender funds. Listed are the different combinations of matching criteria used.

Own	Return Proxy				No. of Observation	Wealth Transfer
	Strategy	Age	Size	Active		
x					3,474	424,829,545
	x	x	x	x	3,092	537,204,555
	x	x	x		3,325	623,628,681
	x	x		x	3,423	611,142,143
	x	x			3,547	917,055,102
	x		x	x	3,585	554,778,422
	x		x		3,595	670,343,299

Table VII Impact on Fund Flow

Intuition: $F_{i,t}/NAV_{i,t}^R$ is the number of shares purchased; the term in parentheses is the overpricing per share.

7.2 Proxying for fair NAV / true return

To construct $NAV_{i,t}^A$, the paper needs a proxy for the "true" return in the month where the fund reports a small gain. It uses random draws from negative returns:

- **Own:** draw from the fund's own negative-return months.
- **Matched Non-offenders:** draw from negative returns of a matched set of Non-offender funds (match on strategy and/or age/size/activity).

Using each proxy, wealth transfers are summed across events, producing Table VII.

7.3 Table VII interpretation

- The estimated wealth transfer in the sample ranges from roughly \$0.4B to \$0.9B depending on the fair-return proxy.
- Scaling by the paper's stated database coverage implies an aggregate industry-wide transfer on the order of \$1B-\$2B over 1994-2005.

Interpretation: even "small" distortions in monthly returns can translate into economically meaningful transfers when flows occur at inflated valuations.

7.4 Key limitations / things to be careful about

- The discontinuity test flags patterns consistent with manipulation but does not observe "true" returns; alternative channels must be continuously scrutinized.
- Holdings/valuation discretion is inferred indirectly through strategy labels, audits, and smoothing proxies.

- Results depend on the quality and coverage of the database and on how missing/zero returns are handled.

8 Conclusion

8.1 Core conclusion

This paper documents a *robust discontinuity at zero* in the pooled distribution of monthly hedge-fund returns: there are “too many” small gains and “too few” small losses relative to what a smooth distribution would predict. This pattern is consistent with the view that at least some hedge fund managers *temporarily overstate returns to avoid reporting losses*.²**Key evidence:**

- The discontinuity appears broadly (including both live and defunct funds), suggesting it is not driven by standard database biases.³
- The discontinuity is *absent* in the three months culminating in an audit, consistent with the idea that tighter oversight reduces the scope for discretionary reporting.
- The discontinuity *disappears* when returns are aggregated to a bimonthly frequency, consistent with the idea that overstatements are later reversed.
- The discontinuity is not reproduced by distributions of fitted returns generated from commonly used hedge-fund risk factors, indicating that it is unlikely to be purely mechanical (e.g., generated by nonlinear strategy payoffs alone).
- The discontinuity is stronger where discretion is more plausible, such as in settings with greater illiquidity or return smoothing.

8.2 Economic significance (what this implies in equilibrium terms)

The paper argues the phenomenon is not merely statistical: it can distort investor beliefs and capital allocation.

- If returns are overstated, reported net asset values (NAVs) are overstated, implying that *new investors can overpay for entry*.
- The authors estimate that a nontrivial fraction of observations may be distorted (on the order of $\sim 10\%$ in their database-based calculation), and the implied aggregate wealth transfer from incoming to existing investors can be economically large when scaled to industry size.
- More broadly, investors may under-estimate downside risk and over-estimate managerial performance, leading to an inefficiently high allocation of capital to some funds.

²The paper emphasizes that the key anomaly is concentrated just below vs just above zero, rather than a general shift in the mean of returns.

³See also the abstract: the discontinuity is present in live and defunct funds, and across ages.

8.3 Why does a kink at zero arise?

A threshold-incentive mechanism plus discretion:

1. **Incentives are discontinuous around zero.** A monthly loss is salient: it can harm marketing narratives, increase redemption risk, and reduce future inflows. Hence there is a strong incentive to avoid reporting small losses, even if the *economic* performance is only slightly negative.
2. **Discretion makes “cosmetic” smoothing feasible.** Hedge funds often hold instruments whose valuation requires judgment (illiquid securities, model-based marks, broker quotes). That discretion can be used to shift a slightly negative true return into a slightly positive reported return.
3. **Overstatements must be unwound.** If a manager inflates returns today, the fund’s true value has not changed. Eventually, subsequent reported performance must be lower (or losses must appear later). The empirical implication is *reversal*, which is consistent with the disappearance of the kink when using bimonthly returns.
4. **Monitoring attenuates manipulation.** Close to audits, the expected cost of misreporting rises (greater scrutiny and reputational/legal risk), so the kink weakens or disappears in audit windows.

8.4 Takeaway

A sharp jump at zero in reported monthly hedge-fund returns is hard to reconcile with skill or strategy payoffs alone, but is consistent with temporary return inflation when valuation discretion is high and oversight is weak, with meaningful implications for investor beliefs and wealth transfers.